



# BIAS IN NETWORKS: Link Prediction

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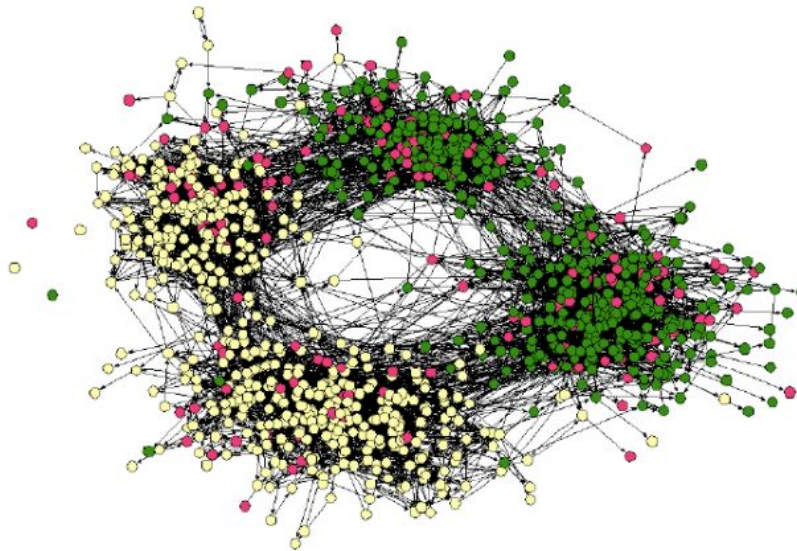
DSCI 531

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# Bias in networks

- Individuals are biased in their preferences for who to link to
  - Homophily (Lazarsfeld and Merton 1954): linked people tend to be more similar than a random pair of individuals



Homophily can organize the network into densely-connected, homogeneous components. The friendship network of high school & middle school students, color shows race, and divisions show high school vs middle school.



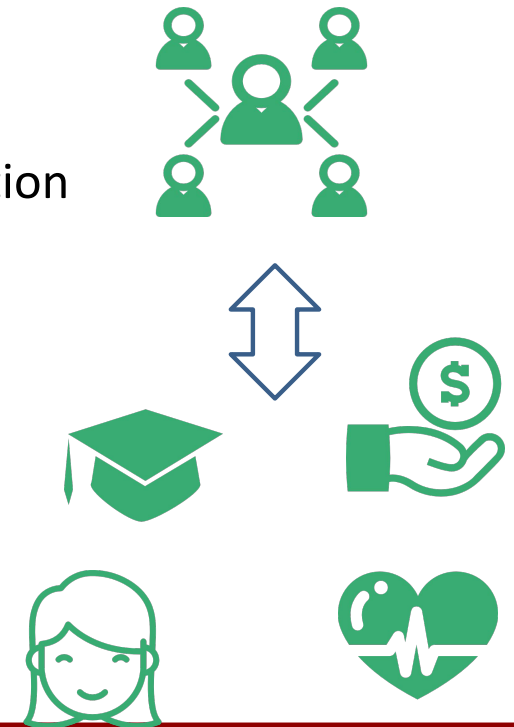
# Bias in Networks

- Individuals are biased in their preferences for who to link to
  - Homophily: linked people are more similar than a random pair of individuals
    - aka “birds of a feather fly together”
  - Assortative mixing
    - E.g., people marry others who are demographically similar
    - Also applies to network structure: popular people more likely to be friends
  - Geometric/spatial constraints on friendships
- Biased preferences shape social networks, with surprising effects



# Some of the consequences of biased linking

- Small world effect
  - Local ↔ Global
- Friendship paradox
  - Friends are a (systematically) biased sample of the population
- “Structural holes” (Burt 1995)
  - Some positions within social networks confer benefits
    - Job and promotion opportunities
    - Novel information
  - Inequality of opportunity
    - Glass ceiling effect
    - Ranking disparity and decreased visibility of minorities



# How large are social networks?

## The Milgram experiment (1963)



- Stanley Milgram asked 160 random people in Kansas and Nebraska to deliver a letter to a stockbroker in Boston.
  - *If you do not know the target, do not try to contact him directly. Instead, mail this letter... to a personal acquaintance who is more likely than you to know the target... you must know this person on a first-name basis.*
- How long would it take?





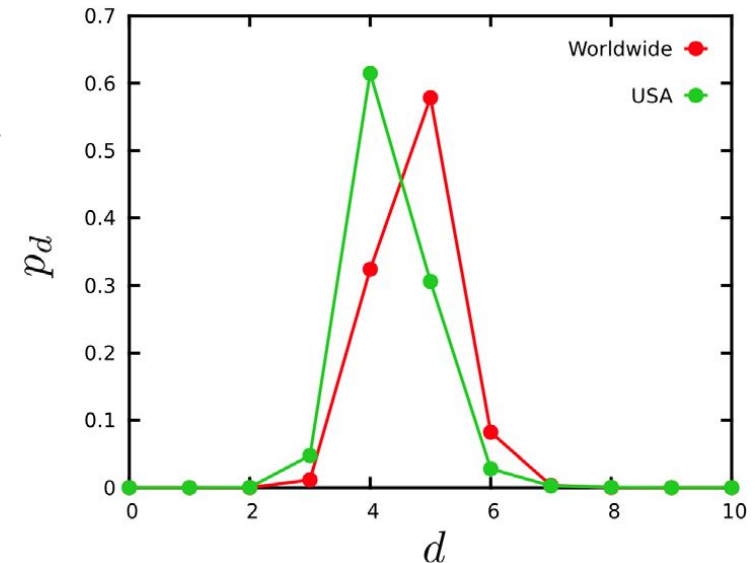
## The Milgram experiment

- Within a few days the first letter arrived, passing through only two links.
- Eventually 44 of the 160 letters made it to the target, some requiring close to a dozen intermediates.
- The median number of steps in completed chains was 5.5
  - “six degrees of separation”



# Facebook is a very small world

- Ugander et al. directly measured distances between nodes in the Facebook social graph (May 2011)
  - 721 million active users
  - 68 billion symmetric friendship links
  - the average distance between the users was 4.74





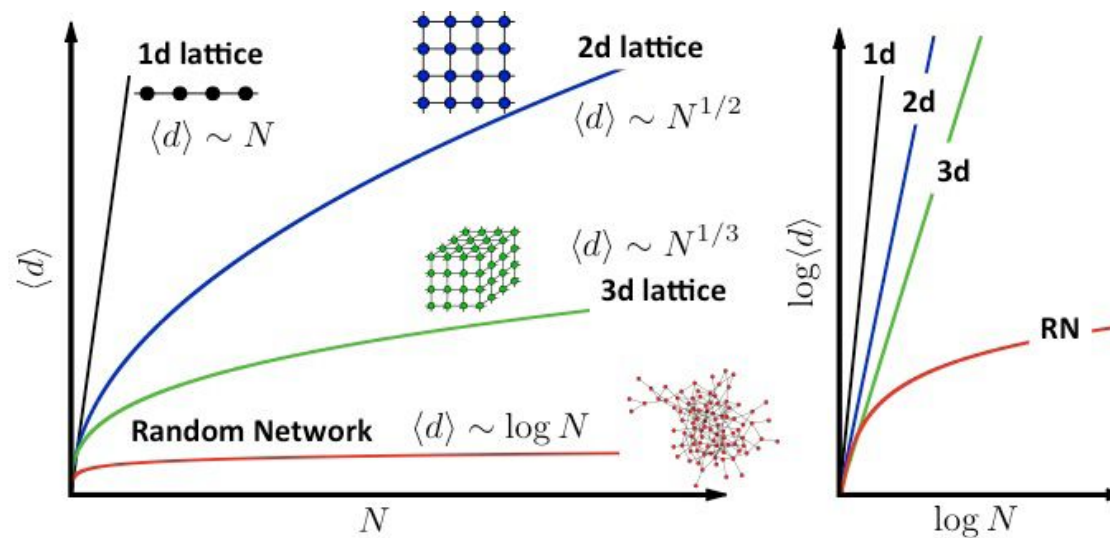
## Is this surprising?

- People do not have **global** knowledge of network structure
  - They only know their own acquaintances (**local** structure)
- Despite this, people were able to create short chains to the target
- Two fundamental aspects of networks
  1. – People are connected by relatively short chains, even over large geographic distances
    - Networks are “small” (small-world). Why is this surprising?
  2. – People are able to find these short chains using local information only
    - Networks are searchable



## Why is it surprising?

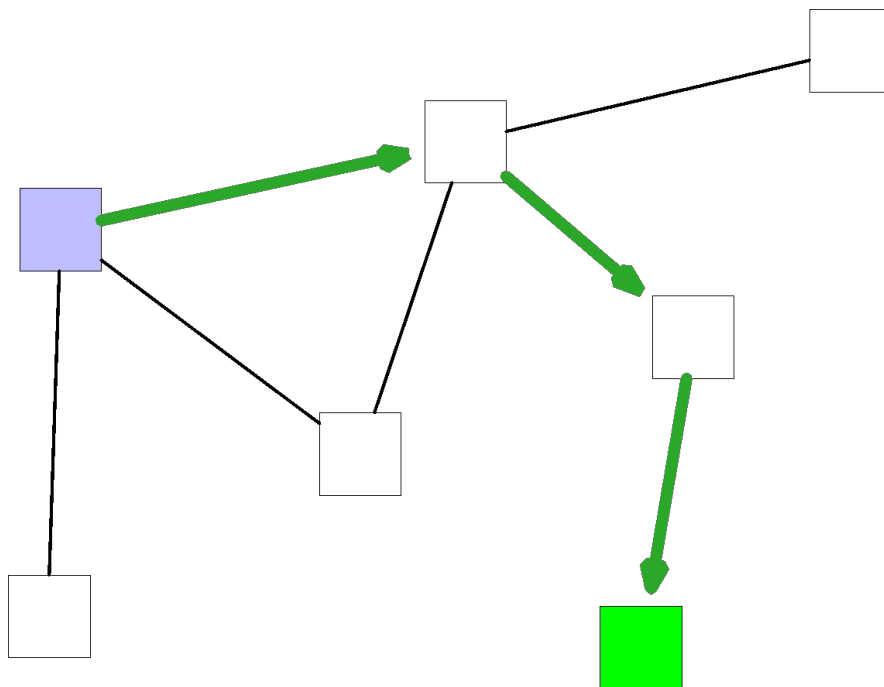
- Regular lattices (e.g., physical geography) do not have the small world property
  - Distances grow polynomially with system size
  - In networks, distances grow logarithmically with network size



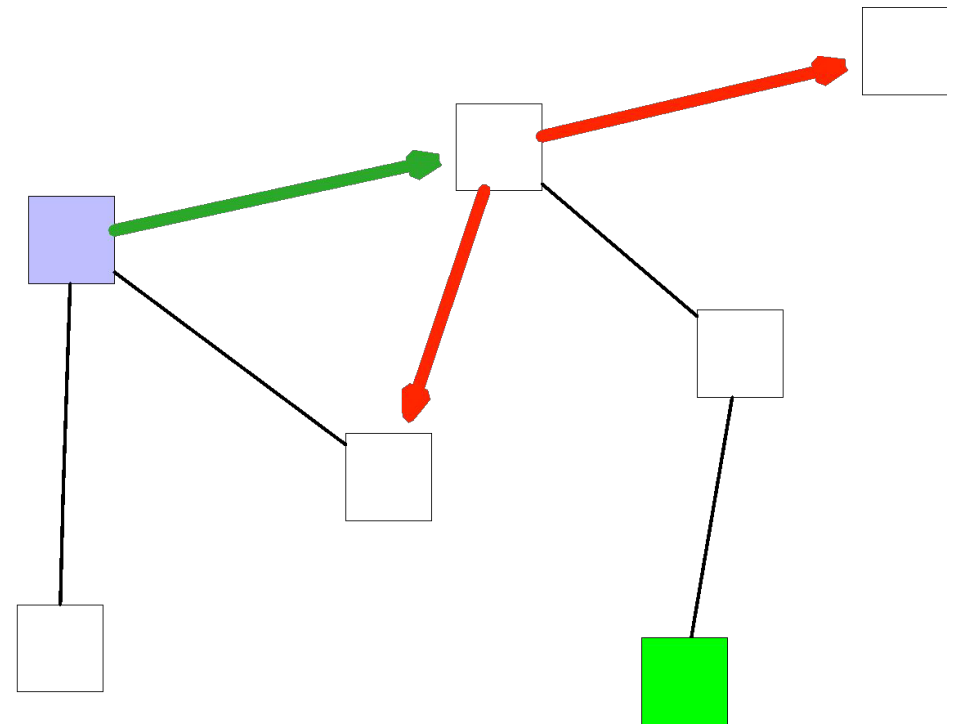


## Implications

Efficiently navigate network



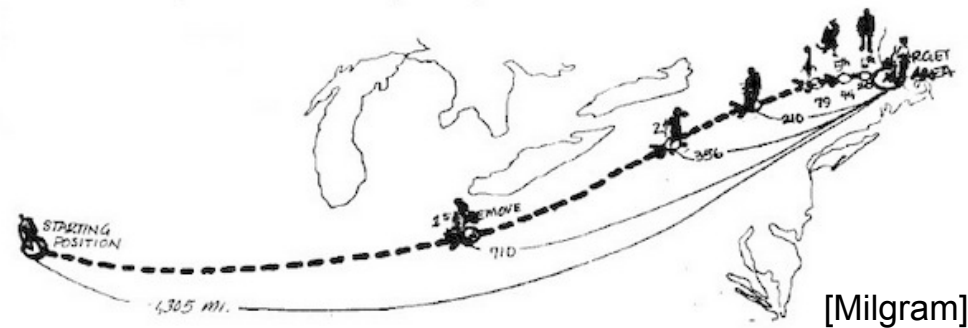
Inefficiently navigate network





# Social networks are searchable

- Milgram experiments showed that
  - Short chains exist!
  - People can find them!
    - Using only local knowledge (who their friends are, their location and profession)
    - How are short chains discovered with this limited information?
    - Hint: geographic information?



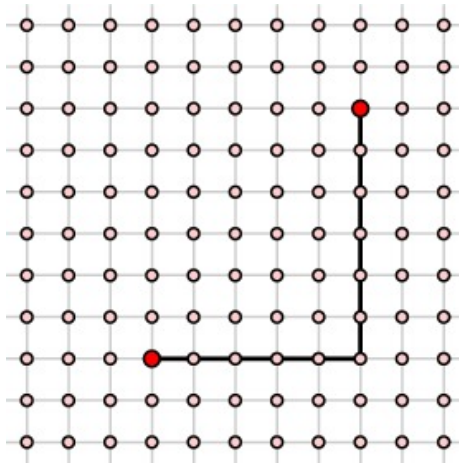
[Milgram]



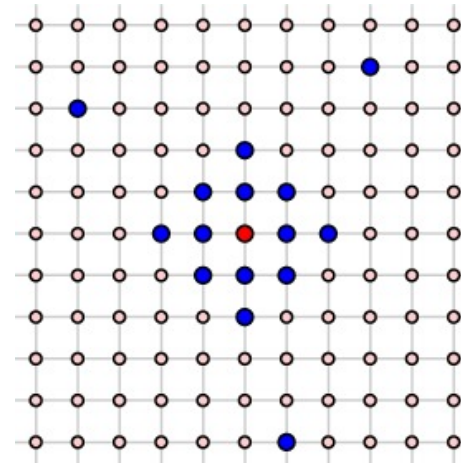
# Kleinberg model of geographic links

- Incorporate geographic distance in the distribution of links

Distance between nodes  
is  $d$



Link to all nodes within distance  $r$ , then  
add  $q$  long range links with probability  $d^{-\alpha}$

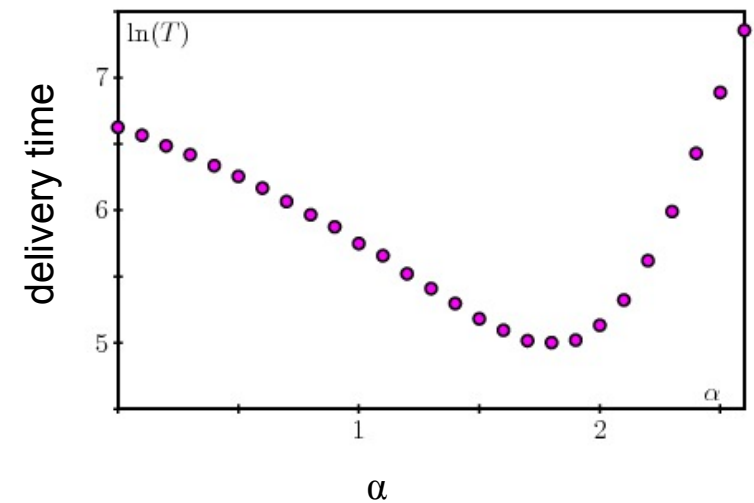


Kleinberg, J. M. (2000). Navigation in a small world. *Nature*, 406(6798), 845-845.



## How does this affect short chains?

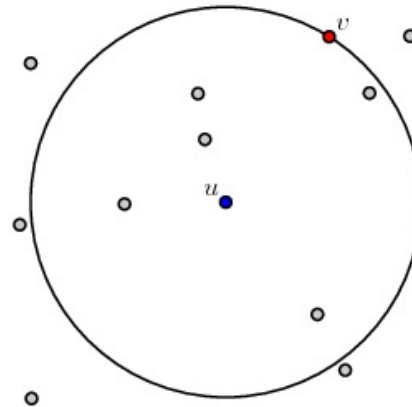
- Simulate Milgram experiment
  - at each time step, a node selects a friend who is closer to the target (in lattice space) and forwards the letter to it
    - Each node uses only local information about its own social network and not the entire structure of the network
  - **Network is searchable via a greedy algorithm when  $\alpha=2$** 
    - i.e., probability to form a link drops as square of distance!
    - Given real networks do not satisfy this result exactly, we assume that real life is more complicated (moreover, this ignores lost messages)





## Does this hold for real networks?

- Liben-Nowell et al. tested Kleinberg's prediction for the LiveJournal network of 1M+ bloggers
  - Blogger's geographic information in profile
  - How does friendship probability in LiveJournal network depend on distance between people?
- People are not uniformly distributed spatially
  - Coasts, cities are denser



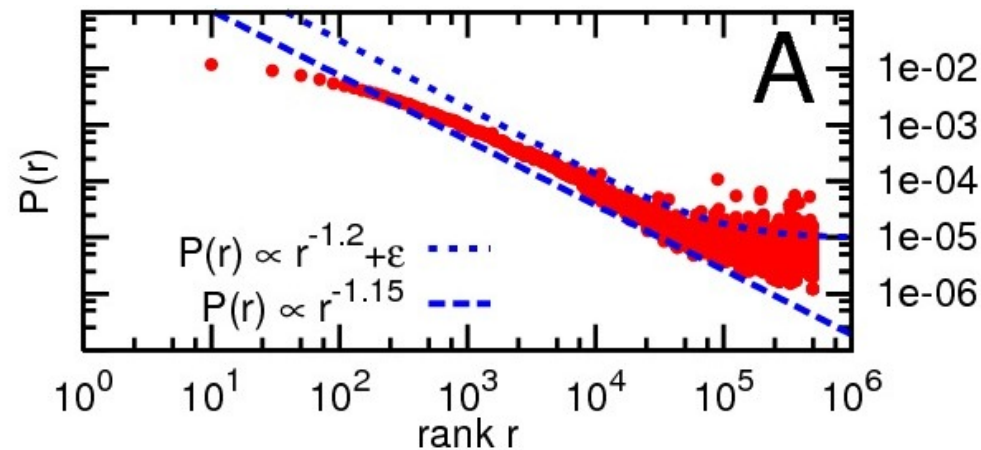
Use rank, instead of distance  $d(u,v)$   
 $\text{rank}_u(v) = 6$

Since  $\text{rank}_u(v) \sim d(u,v)^2$ , and link probability  
 $\Pr(u \square v) \sim d(u,v)^{-2}$ ,  
we expect that  $\Pr(u \square v) \sim 1/\text{rank}_u(v)$



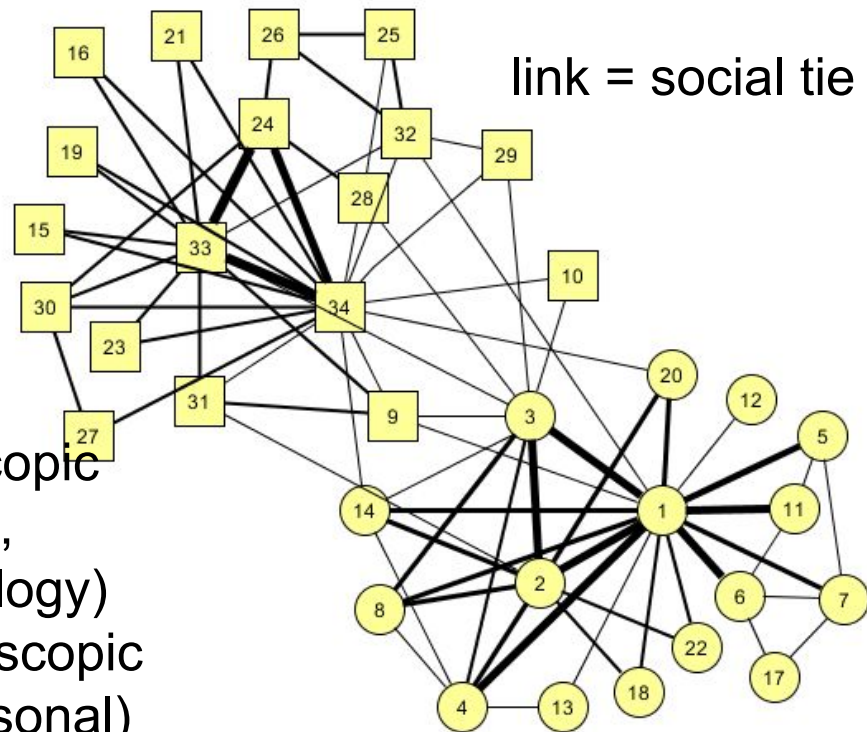
# LiveJournal is a searchable network

- Probability that a link exists between two people as a function of the distance rank between them
  - LiveJournal is a rank-based network  $\Rightarrow$  it is searchable





# Not all social ties are created equal



Can macroscopic structure (i.e., network topology) predict microscopic (i.e., interpersonal) relationship?

How does microscopic structure (i.e., strength of social ties) affect macroscopic phenomena (e.g., spread of information)?



# The Strength of Weak Ties<sup>1</sup>

Mark S. Granovetter

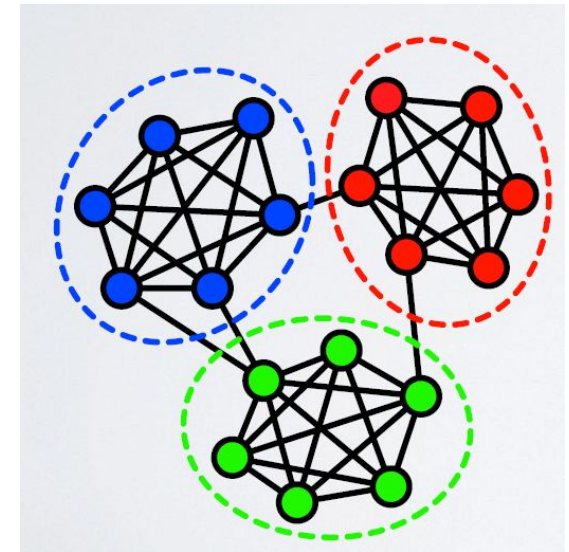
*Johns Hopkins University*

novel information comes from weak ties

- within-community links = strong
- bridge links = weak

**why?**

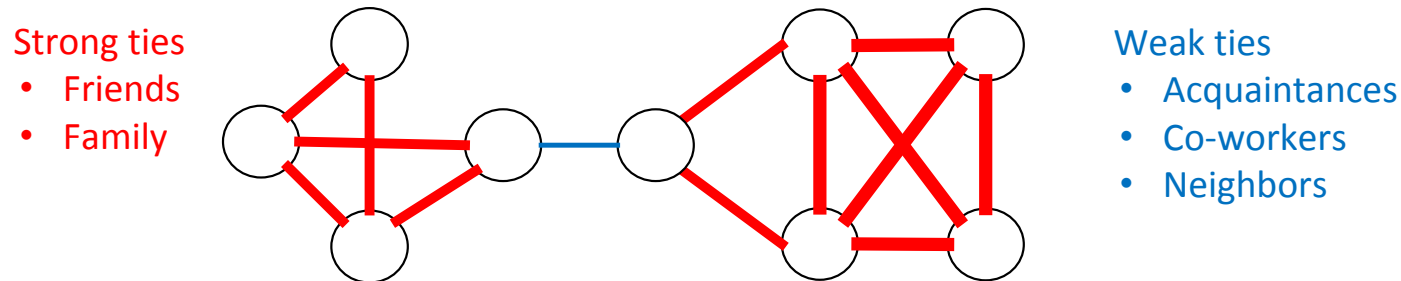
- information propagates *quickly within* a community,
- but *slowly between* communities





# Social ties

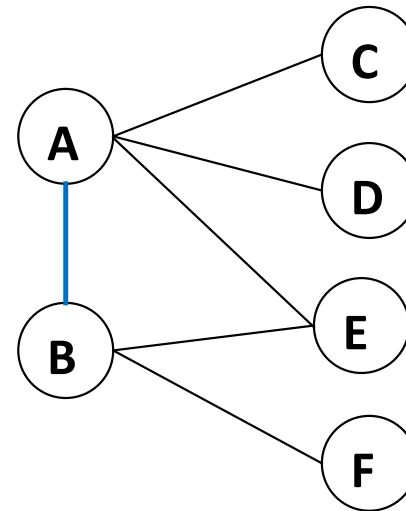
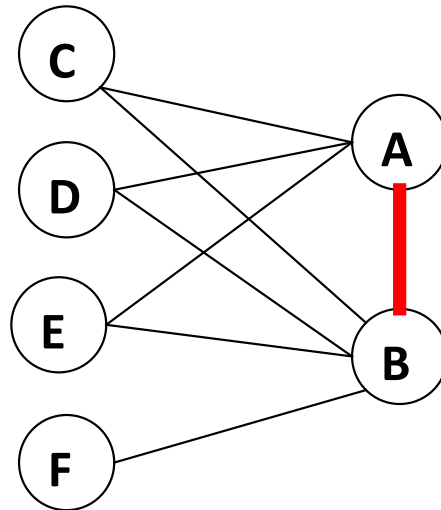
- Strength of a tie between two people is characterized by
  - Frequency and length of interactions
  - Emotional intensity of interactions
  - Intimacy, ...
- **Strong ties** vs **weak ties**





## Measuring tie strength

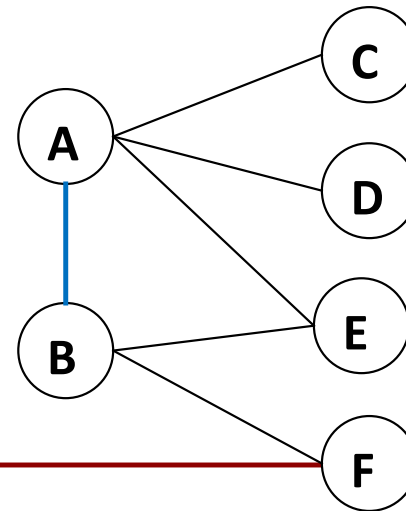
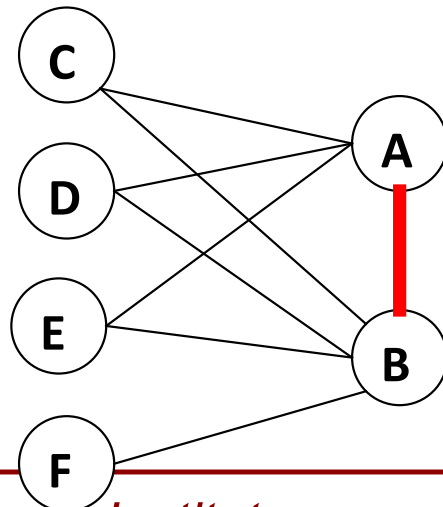
- Intensity and duration of social interactions is reflected in network structure
- Consider nodes A and B
  - Set  $S=\{C, D, E, \dots\}$  of all nodes tied to *either* or *both* A & B
- Hypothesis: the stronger the tie between A & B, the larger the number of nodes in S to whom both A & B are tied





## Measuring tie strength

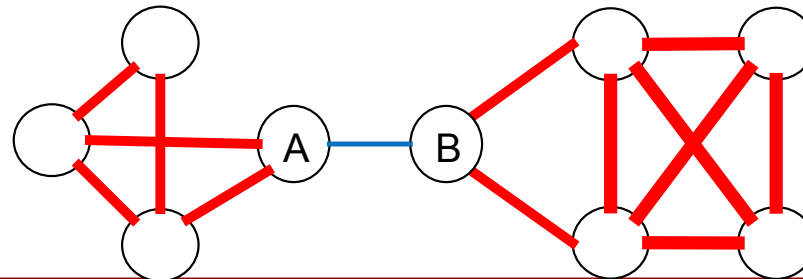
- Strong ties: large overlap in friendship circles
  - These people spend much time together
- Weak ties: small overlap in friendship circles
  - These people occasionally spend time together
- No ties: no overlap in friendship circles





## A “bridge”

- **Bridge**
  - a line in a network which provides the only path between two nodes, e.g., A-B
  - Link A-B provides the only route for information and influence to flow between communities
- **All bridges are weak ties**
  - Weak ties play an important role in diffusion

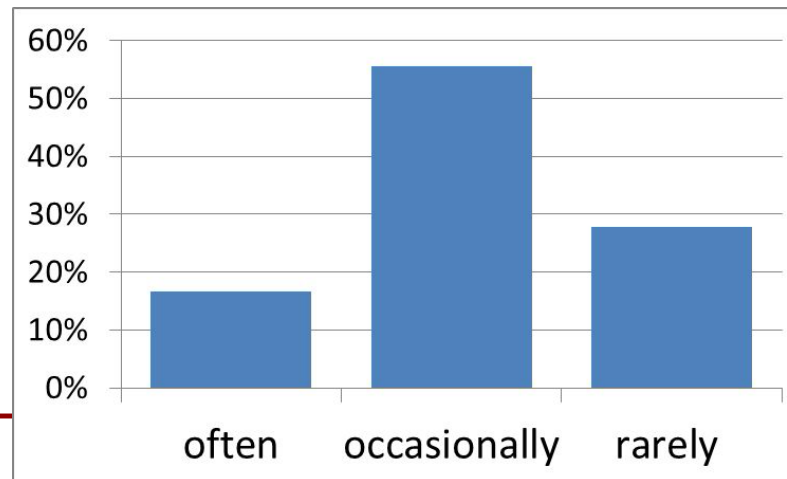




# Weak ties & novel information: Job search study

- Workers find out about new jobs through personal contacts
- How does worker-contact tie strength affect chances of learning about the new job?
  - Measure tie strength by frequency of contact
    - Often (at least once a week)
    - Occasionally (less than 2x/week, more than 1x/year)
    - Rarely (1x/year or less)

What kind of contacts helped workers (N=54) find jobs?

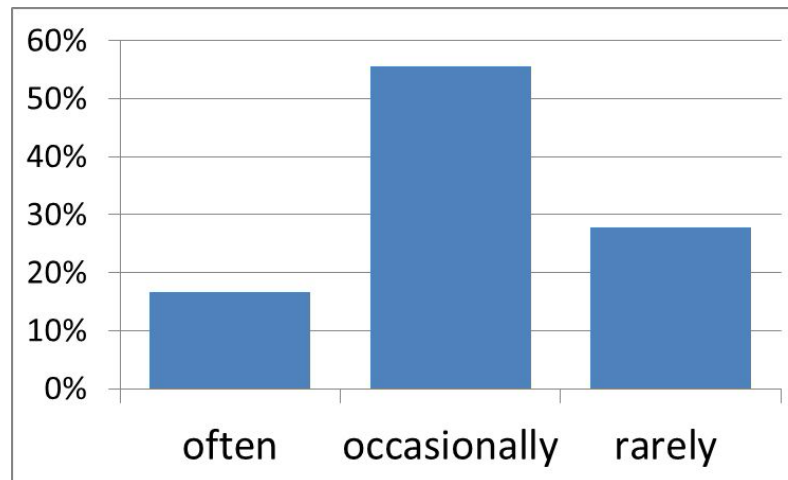




## Job search study

- Chance encounters with weak ties (college friends, former workmates) provided most of the job leads
  - “It is remarkable that people receive crucial information from individuals whose very existence they have forgotten”

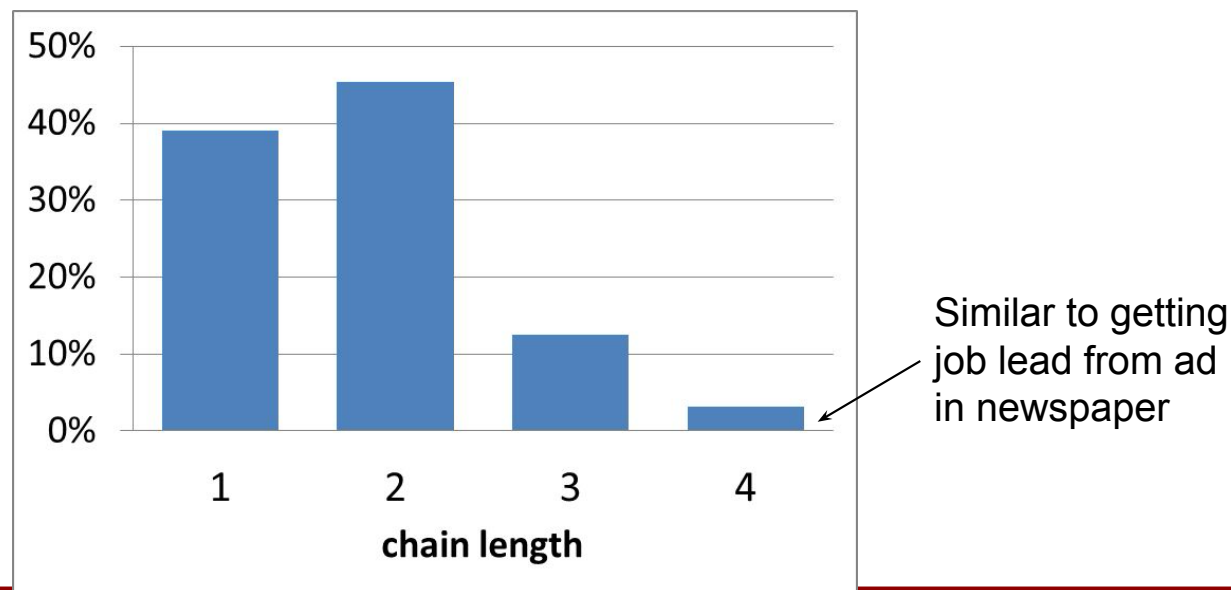
What kind of contacts helped workers (N=54) find jobs?





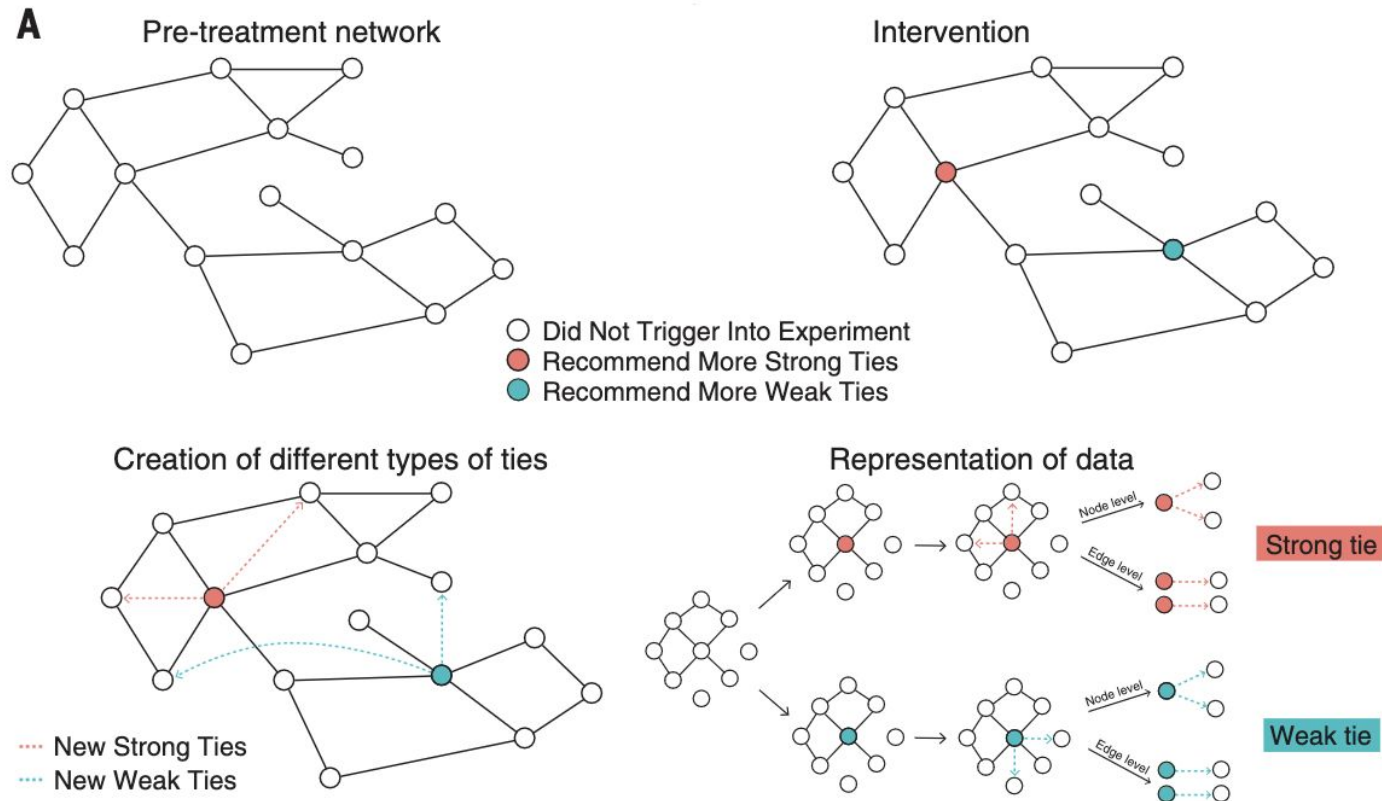
## Length of chains

- **Granovetter** traced job lead back to its source, i.e., where did the contact get information about the job from?
  - direct lead from the employer → length 1
  - Lead from a single intermediary → length 2





# Further proof

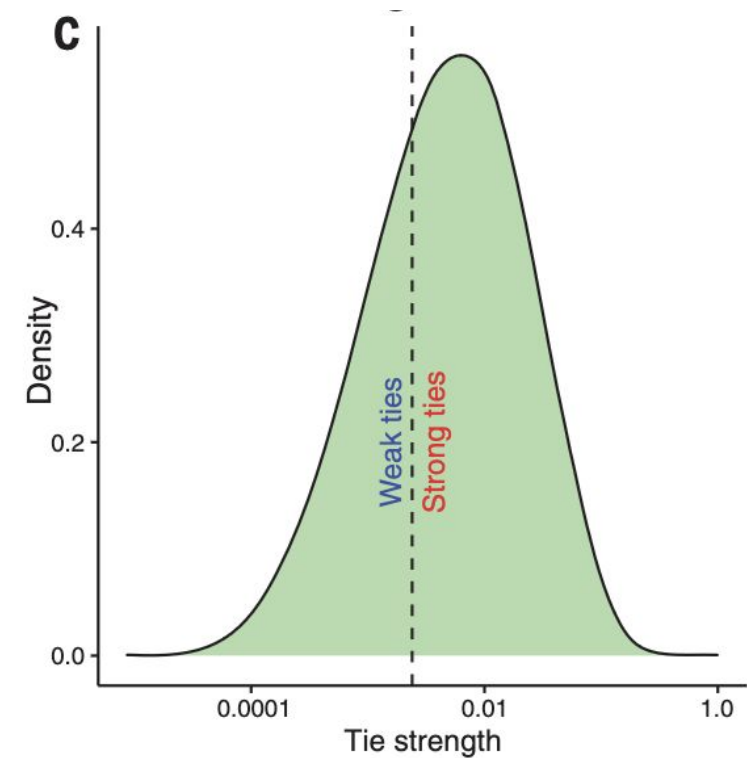


Rajkumar et al., 2022



## Weak ties based on tie strength

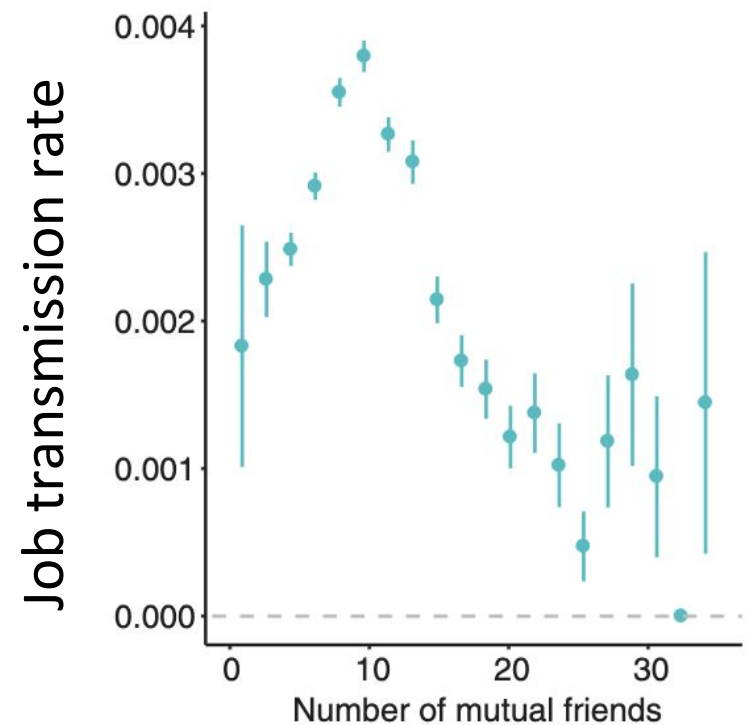
(If users have few friends in common they have weak ties)



## Strength of weak ties

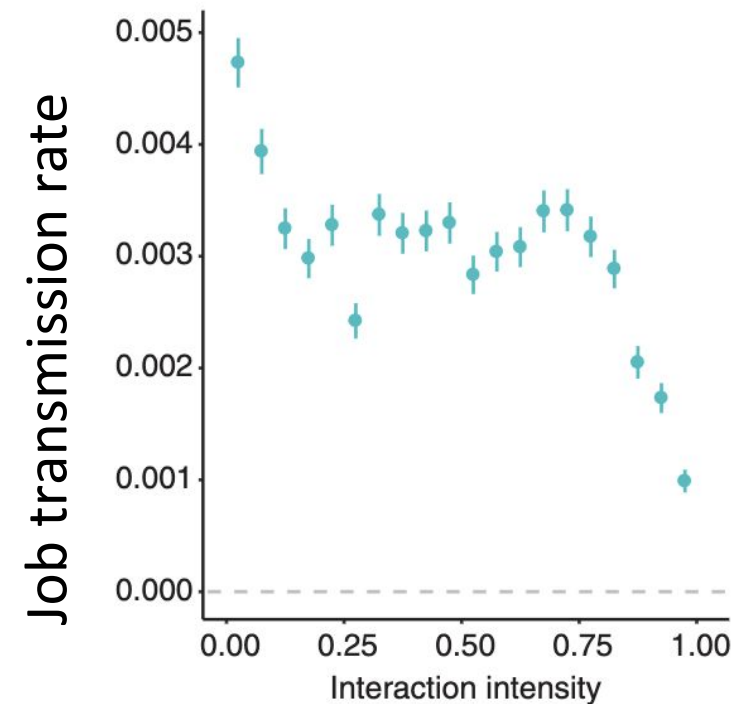


Results show weak ties increase job likelihood based on interaction intensity...





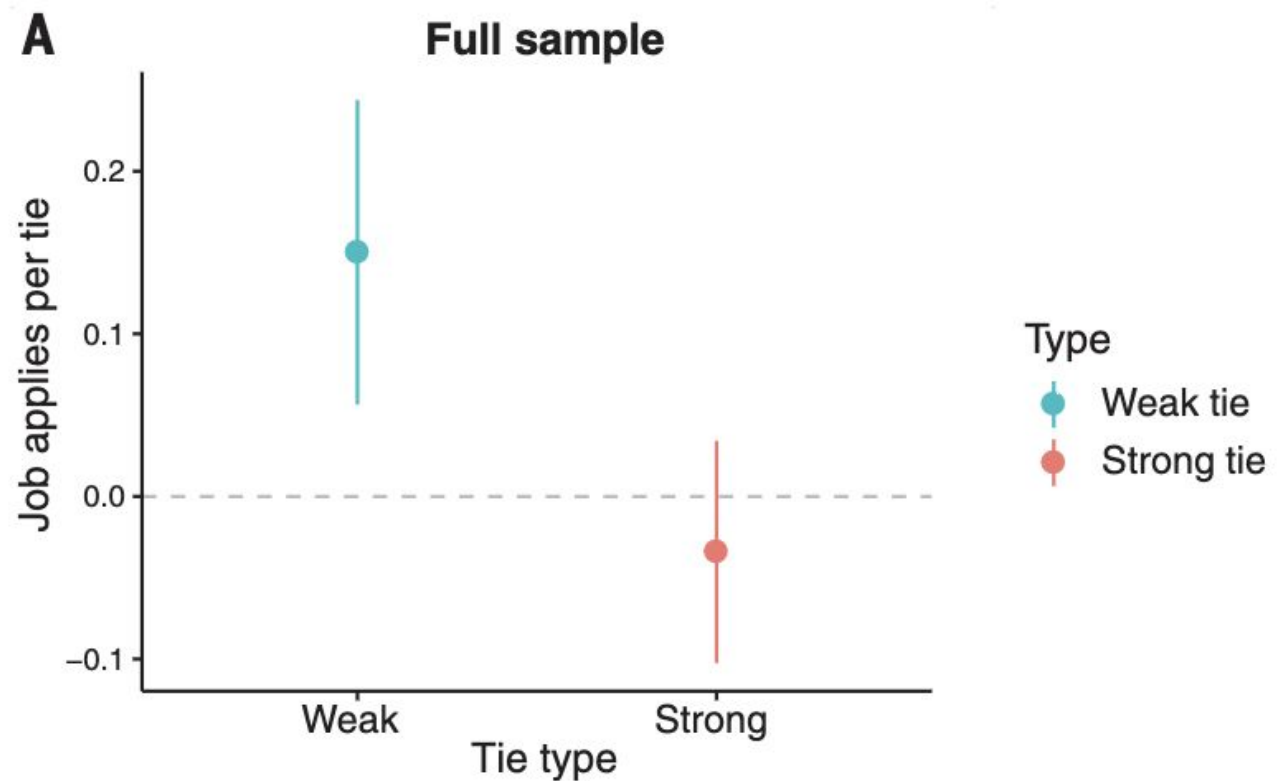
We see similar results for many mutual friends





# Weak ties matter!

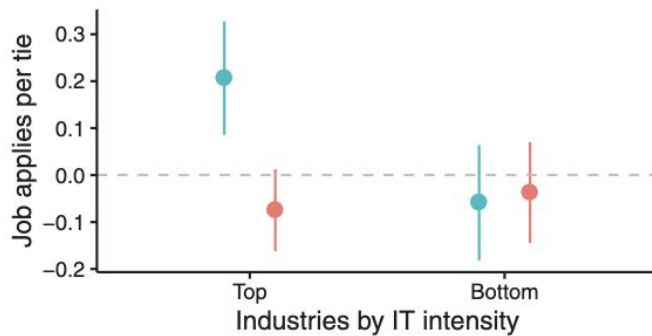
Weak ties increase the chance of a job



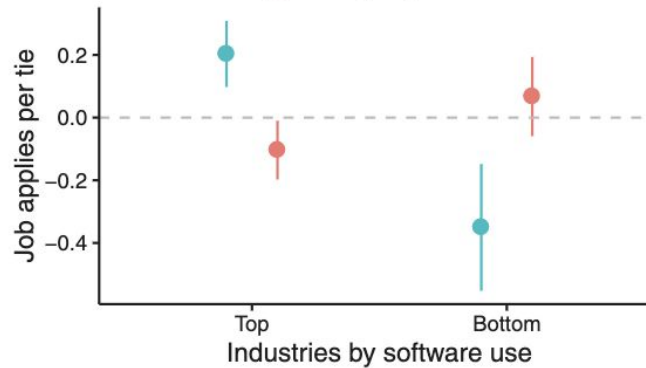


# Effect varies by industry

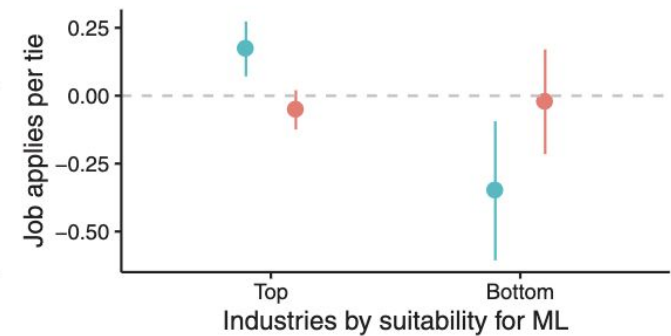
**B** Heterogeneity by information technology



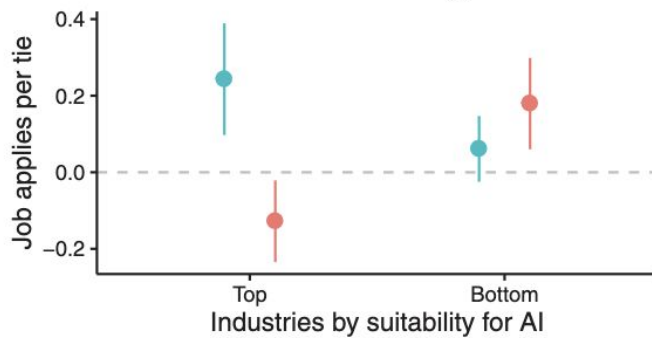
**C** Heterogeneity by software use



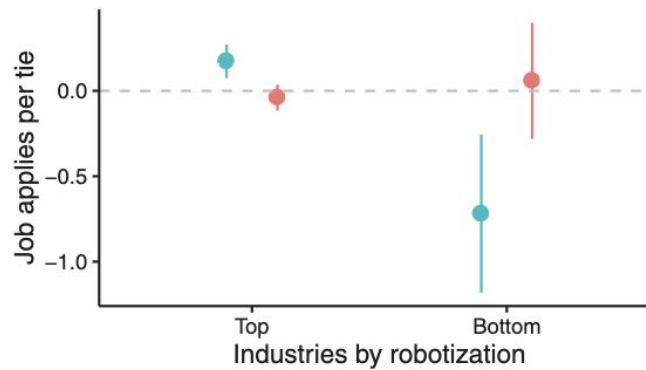
**D** Heterogeneity by suitability for machine learning



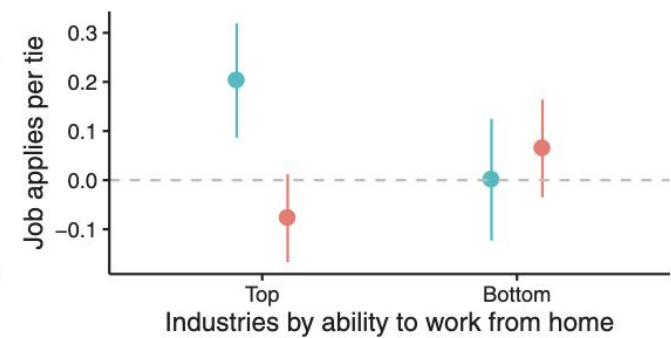
**E** Heterogeneity by suitability for artificial intelligence



**F** Heterogeneity by robotization



**G** Heterogeneity by ability to work from home





## Strength of social ties (review)

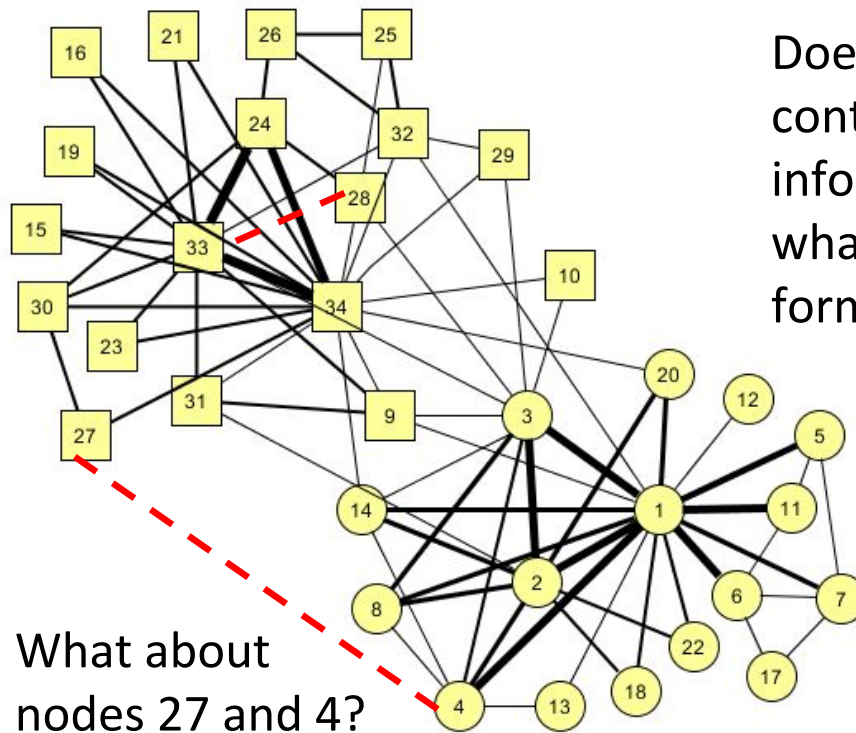
- Strong ties
  - surrounded by *many* mutual friends
  - characterized by lots of shared time together
- Weak ties
  - have *few* mutual friends
  - Serve as bridges to diverse parts of the network
  - Provide access to novel information



# Link Prediction

Will nodes 33 and 28 become friends?

Does network structure contain enough information to predict what new links will form in the future?



What about nodes 27 and 4?



# Link insertion & Web navigation

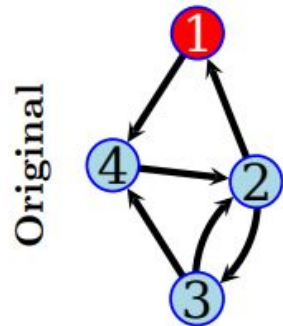


Help make web sites more searchable

- Make it easier to find articles on Wikipedia
- Direct traffic to overlooked products on online shopping Web sites



# Link insertion and Web navigation



Adjacency matrix

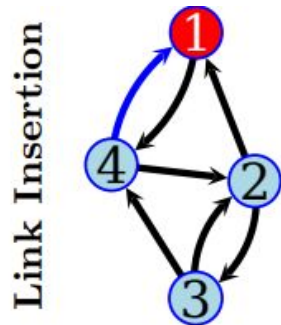
$$W = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

$$P = \begin{pmatrix} 0.0 & 0.5 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.5 & 1.0 \\ 0.0 & 0.5 & 0.0 & 0.0 \\ 1.0 & 0.0 & 0.5 & 0.0 \end{pmatrix}$$

PageRank

$$\pi = \begin{pmatrix} \mathbf{0.18} \\ 0.36 \\ 0.18 \\ 0.27 \end{pmatrix}$$

Insert a link to bias navigation toward a certain page



$$W' = \begin{pmatrix} 0 & 1 & 0 & \mathbf{1} \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

$$P' = \begin{pmatrix} 0.0 & 0.5 & 0.0 & \mathbf{0.5} \\ 0.0 & 0.0 & 0.5 & \mathbf{0.5} \\ 0.0 & 0.5 & 0.0 & 0.0 \\ 1.0 & 0.0 & 0.5 & 0.0 \end{pmatrix}$$

$$\pi' = \begin{pmatrix} \mathbf{0.22} \\ 0.33 \\ 0.17 \\ 0.28 \end{pmatrix}$$

F Geigl, K Lerman, S Walk, M Strohmaier, D Helic [Assessing the navigational effects of click biases and link insertion on the web](#) - In Hypertext, 2016



# Item Recommendation

## Customers Who Viewed This Item Also Viewed



Logitech Professional  
Presenter R800 with Green  
Laser Pointer

★★★★☆ 1,301

\$60.78 ✓Prime



Kensington Wireless  
Presenter with Laser  
Pointer

★★★★☆ 1,151

\$29.99 ✓Prime

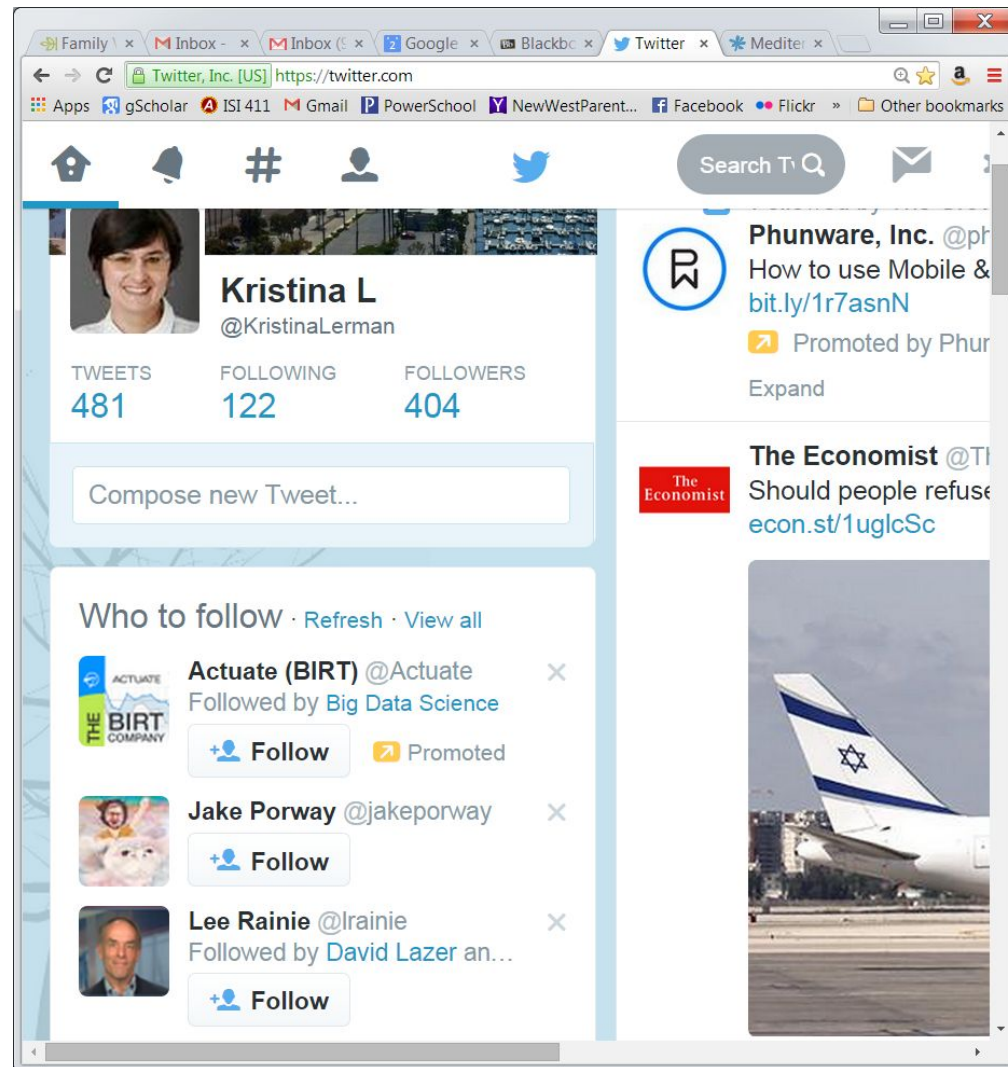


Clever Wireless Presenter  
C748 With Pointer

★★★★☆ 155

\$16.99 ✓Prime

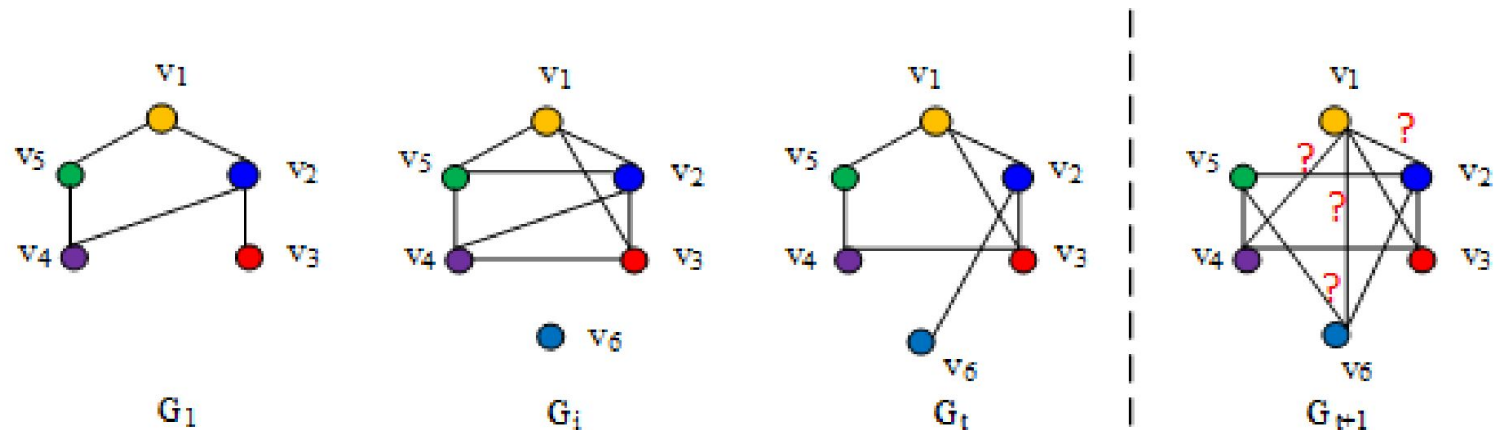
# Who to follow





# Link Prediction

Given a snapshot of a network, infer which new interactions between nodes are likely to occur in the future





# Methods for Link Prediction

- Graph heuristics based on measures for analyzing the “proximity” of nodes in a network.
- Information Flow
- Name visibility and attention
- Graph embedding
- Hybrid approaches

# The Link-Prediction Problem for Social Networks (Liben-Nowell & Kleinberg)



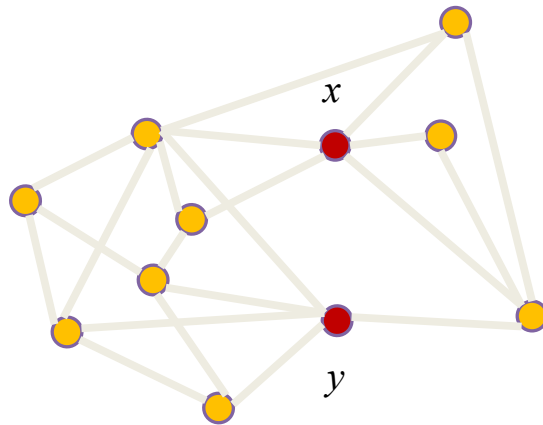
To what extent can the evolution of a social network be modeled using features *intrinsic to the network itself*?

- Formalize the link prediction problem
  - Given a snapshot of a network, infer which new interactions between nodes are likely to occur in the future
- Propose link **prediction** heuristics based on measures for analyzing the “**proximity**” of nodes in a network.
- **Evaluate** link prediction heuristics on large coauthorship networks.  
Future co-authorships can be extracted from network topology.

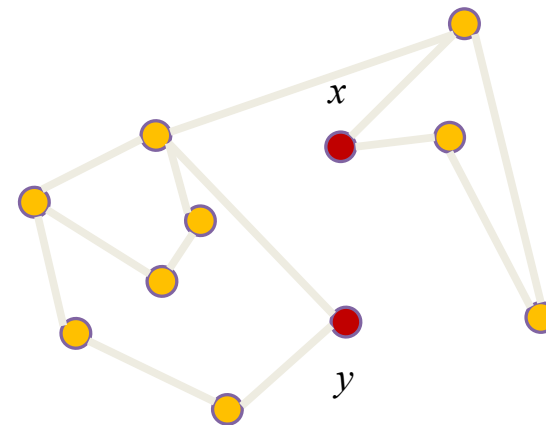


## The intuition

- In many networks, people who are “close” belong to the same social circles and will inevitably encounter one another and become linked themselves.
- Graph heuristics measure how “close” people are



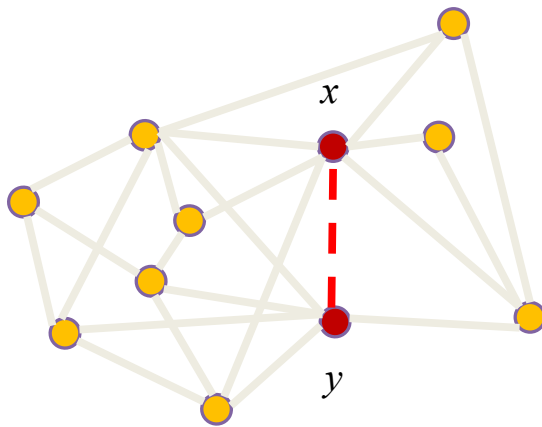
Red nodes are close to each other



Red nodes are more distant



# Link prediction heuristics



- Local

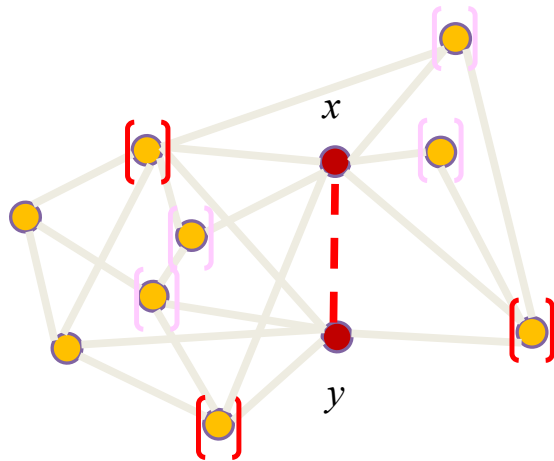
- Common neighbors (CN)
- Jaccard (JC)
- Adamic-Adar (AA)
- Preferential attachment (PA) ...

- Global

- Katz score
- Hitting time
- PageRank ...



# Local link prediction heuristics

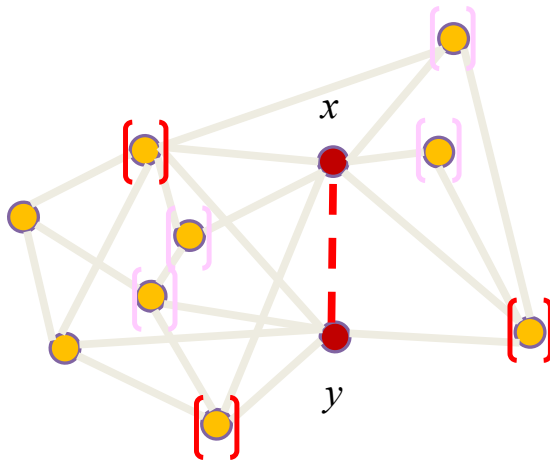


$$CN = 3$$

- Link prediction heuristics
  - Common neighbors (CN)
    - Neighborhood overlap
  - Jaccard (JC)
  - Adamic-Adar (AA)
  - Preferential attachment (PA)



# Local link prediction heuristics

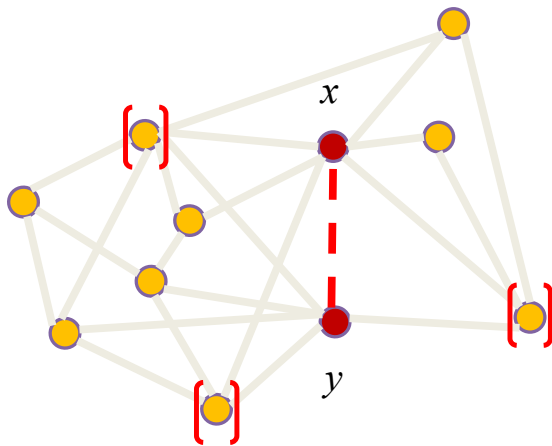


- Link prediction heuristics
  - Common neighbors (CN)
  - Jaccard (JC)
    - Fraction of common neighbors
  - Adamic-Adar (AA)
  - Preferential attachment (PA)

$$JC = \frac{CN}{d_x + d_y - CN}$$



# Link prediction heuristics

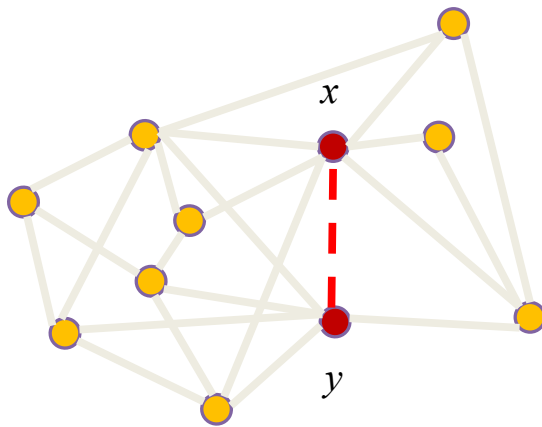


- Link prediction heuristics
  - Common neighbors (CN)
  - Jaccard (JC)
  - Adamic-Adar (AA)
    - Nmbr common neighbors, with each neighbor  $z$  attenuated by log of its degree
  - Preferential attachment (PA)

$$AA = \sum_{z \in CN} \frac{1}{\log d_z}$$



# Local link prediction heuristics

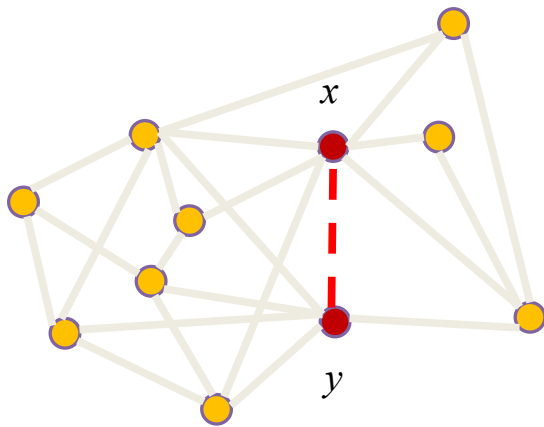


- Link prediction heuristics
  - Common neighbors (CN)
  - Jaccard (JC)
  - Adamic-Adar (AA)
  - Preferential attachment (PA)
    - Better connected nodes are more likely to form more links

$$PA = d_x d_y$$



# Global link prediction heuristics



- Link prediction heuristics

- Katz score

- Measures number of paths between two nodes, attenuated by their length

- Hitting time

- Expected time for a random walk from  $x$  to reach  $y$

- ...



# Which heuristic is the best?



# Data

- Collaboration networks of physicists
  - Core nodes: authors who published at least 3 papers during the training period and at least 3 papers during test period
- Training data: graph  $G(t_0, t_0')$  of collaborations during time period  $[t_0, t_0']$  with  $V$  core nodes and  $E_{old}$  edges
- Test data: graph  $G(t_1, t_1')$  of collaborations during a later time period  $[t_1, t_1']$  with  $V$  core nodes and  $E_{new}$  edges

|          | Training Period |          |                             | Core    |           |           |
|----------|-----------------|----------|-----------------------------|---------|-----------|-----------|
|          | Authors         | Articles | Collaborations <sup>a</sup> | Authors | $E_{old}$ | $E_{new}$ |
| astro-ph | 5,343           | 5,816    | 41,852                      | 1,561   | 6,178     | 5,751     |
| cond-mat | 5,469           | 6,700    | 19,881                      | 1,253   | 1,899     | 1,150     |
| gr-qc    | 2,122           | 3,287    | 5,724                       | 486     | 519       | 400       |
| hep-ph   | 5,414           | 10,254   | 47,806                      | 1,790   | 6,654     | 3,294     |
| hep-th   | 5,241           | 9,498    | 15,842                      | 1,438   | 2,311     | 1,576     |



# Evaluation metric

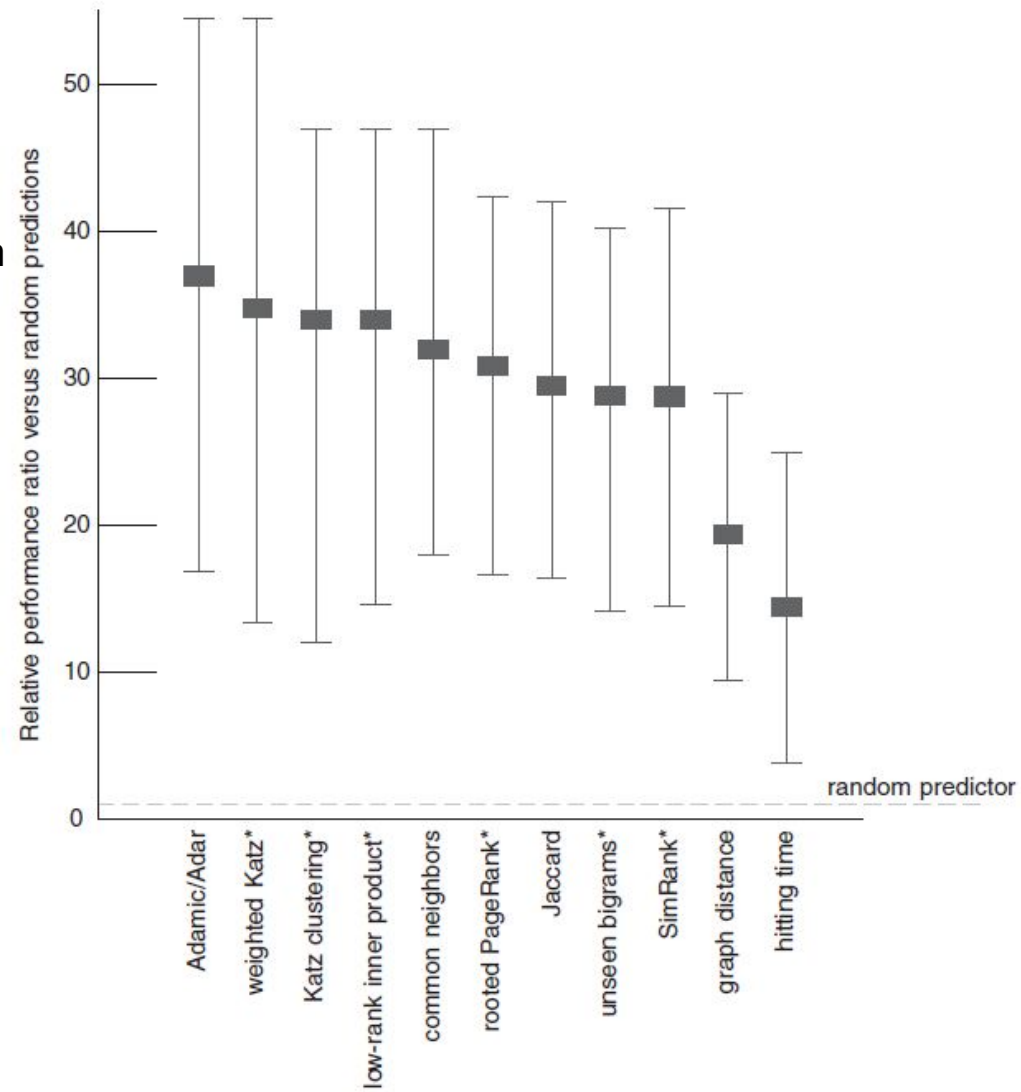
- Link prediction algorithm

- Score node pairs using a heuristic  $p$
- New links more likely among high scoring pairs

- Each link prediction heuristic  $p$  outputs a ranked list  $L$  of new collaborations: pairs in  $V \times V - E_{old}$ .
- Focus evaluation on new links  $E_{new}^*$  between core nodes
- Performance metric: How many of the top  $n$  pairs in ranked list  $L$  are the actual new nodes in  $E_{new}^*$ ?



## Heuristics vs random predictor



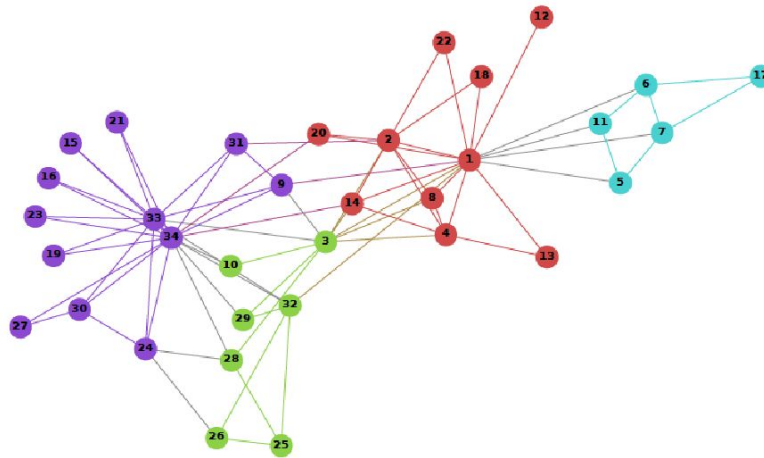


# NODE AND GRAPH EMBEDDING

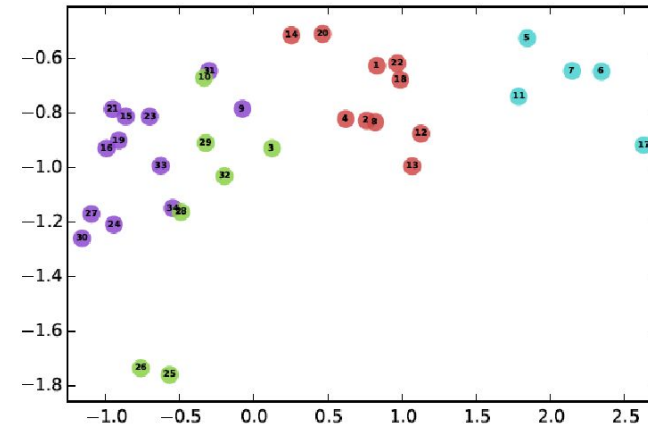
Slides courtesy of Leskovec, Hamilton, Sosc



# Node Embeddings



Input



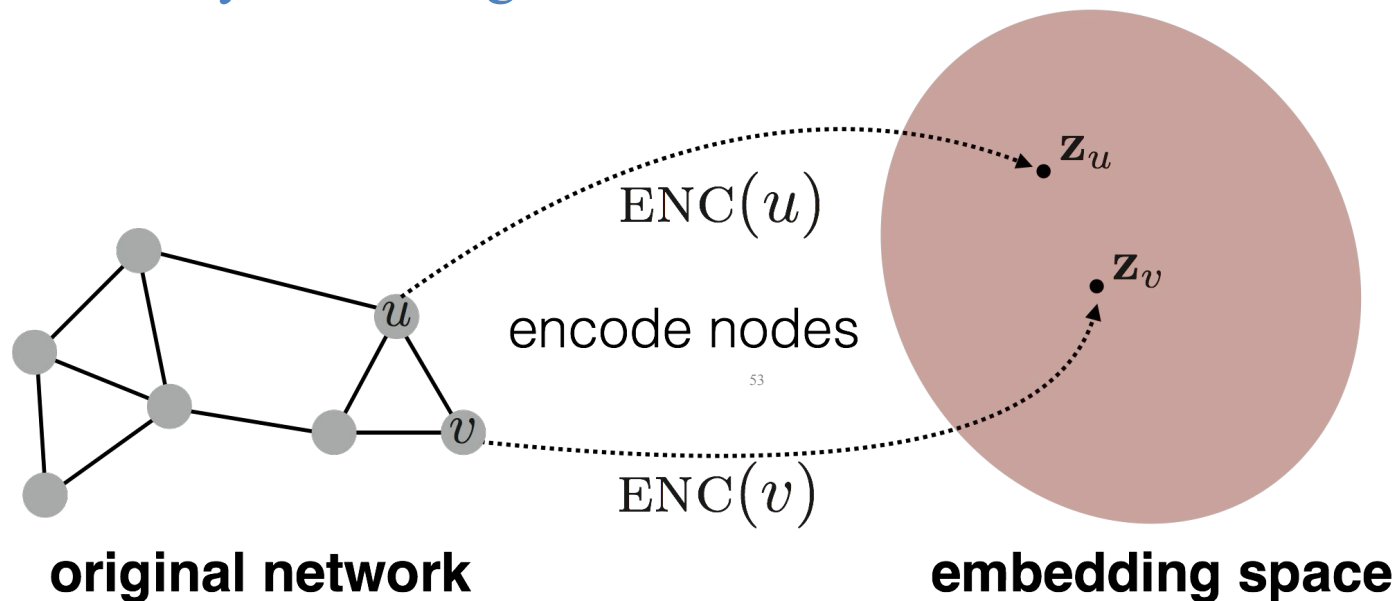
Output

**Intuition:** Find embedding of nodes to d-dimensions so that “similar” nodes in the graph have embeddings that are close together.  
*cf* Graph as text, and nodes as words.



# Embedding Nodes

- Goal is to encode nodes so that **similarity in the embedding space (e.g., dot product)** approximates **similarity in the original network**.





# Learning Node Embeddings

1. **Define an encoder** (i.e., a mapping from nodes to embeddings)
2. **Define a node similarity function** (i.e., a measure of similarity in the original network).
3. **Optimize the parameters of the encoder so that:**

$$\text{similarity}(u, v) \approx \mathbf{z}_v^\top \mathbf{z}_u$$



# Random-walk Embeddings

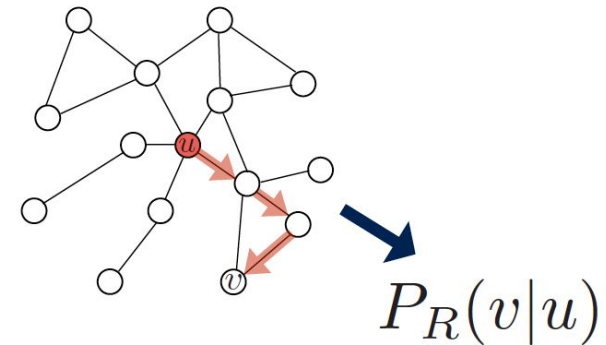
$$\mathbf{z}_u^\top \mathbf{z}_v \approx$$

probability that  $u$  and  $v$   
co-occur on a random  
walk over the network

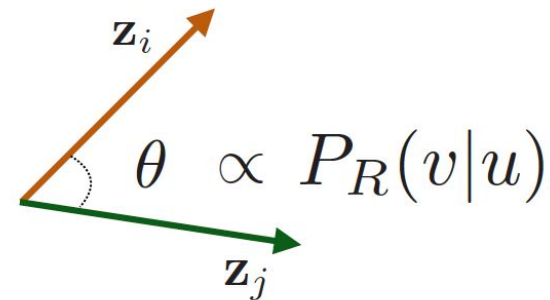


# Random-walk Embeddings

1. Estimate probability of visiting node  $v$  on a random walk starting from node  $u$  using some random walk strategy  $R$ .



2. Optimize embeddings to encode these random walk statistics.



# How should we walk on a graph?



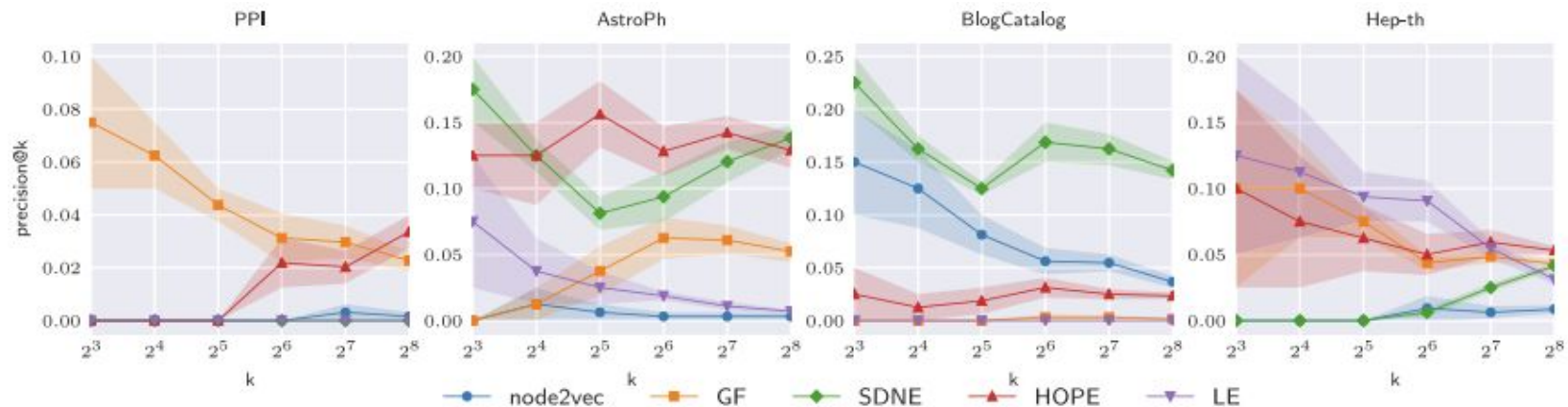
What strategies should we use to run these random walks?

- DeepWalk: runs fixed-length random walks starting from each node. Preserve higher-order proximity between nodes by maximizing the probability of observing the last  $k$  nodes and the next  $k$  nodes in the random walk centered at node.
- Node2vec: In contrast to DeepWalk, node2vec employs biased random walks that provide a trade-off between breadth-first and depth-first graph searches, and hence produces more informative embeddings than DeepWalk.



# Link Prediction with Embeddings

- Hide 20% of the links in a graph and attempt to reconstruct them.
- Rank pairs of node according to similarity (high similarity means they should be linked)
- Precision of top k predicted links
- Some methods perform better than others depending on data



<https://pdf.sciencedirectassets.com/271505/1-s2.0-S0950705118X00112/1-s2.0-S0950705118301540/main.pdf>

PPI: Protein-Protein Interaction networks



# Fairness

# Algorithmic Glass Ceiling in Social Networks

The effects of social recommendations on network diversity



Ana-Andreea Stoica

Christopher Riederer

Augustin Chaintreau

- algorithmic glass ceiling: link recommendation algorithms affect degree distributions and decrease female representation
- Mathematical analysis of conditions creating the glass ceiling effect
- Propose a recommendation algorithm that mitigates some of the biases



## Data collection

- Crawl Instagram's public profiles
- Through the Instagram API, collect profile information and recursively followed the lists of followers.
  - Profile name, username, unique id,
  - Photo meta-data: unique photo id, number of likes, number of comments, hashtags, location, timestamp, tagged users
- In total, 115, 796, 284 photos from 245, 038 different users over multiple months of 2014 and 2015.
- Labeled genders of 92, 935 users: 53%, female and 47% male



# Instagram activity graph

- Instagram activity graph
  - 1 million randomly sampled photos with 3-5 comments sampled from them
  - Create an edge between photo's author and users who comment on it
  - 553,628 nodes and 652,931 edges,
  - 54% female
- DBLP graph
  - Co-authorship graph
  - 102, 263 nodes and 199, 679 edges,
  - 19.3% female

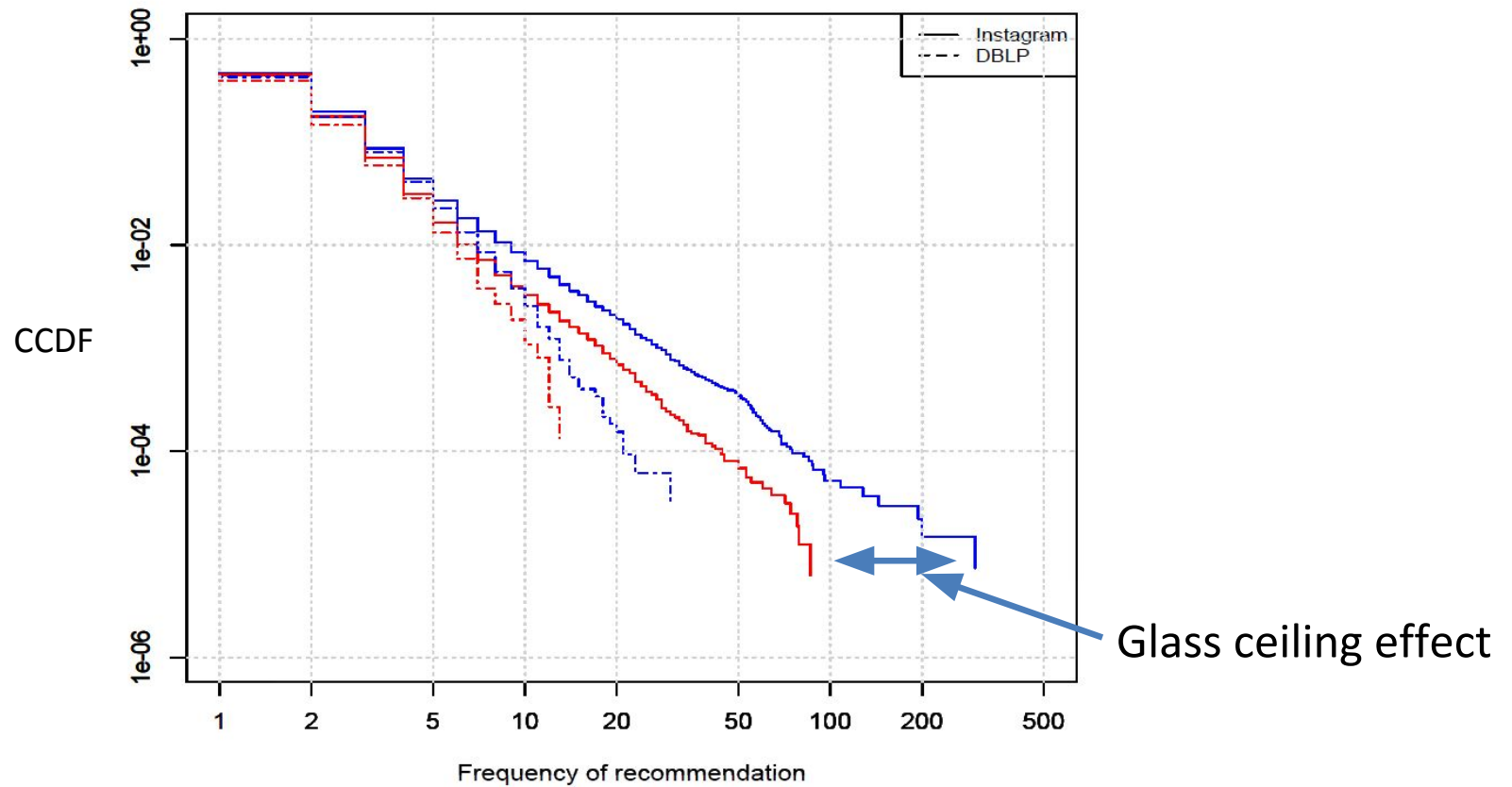


# Organic growth vs growth by recommendation

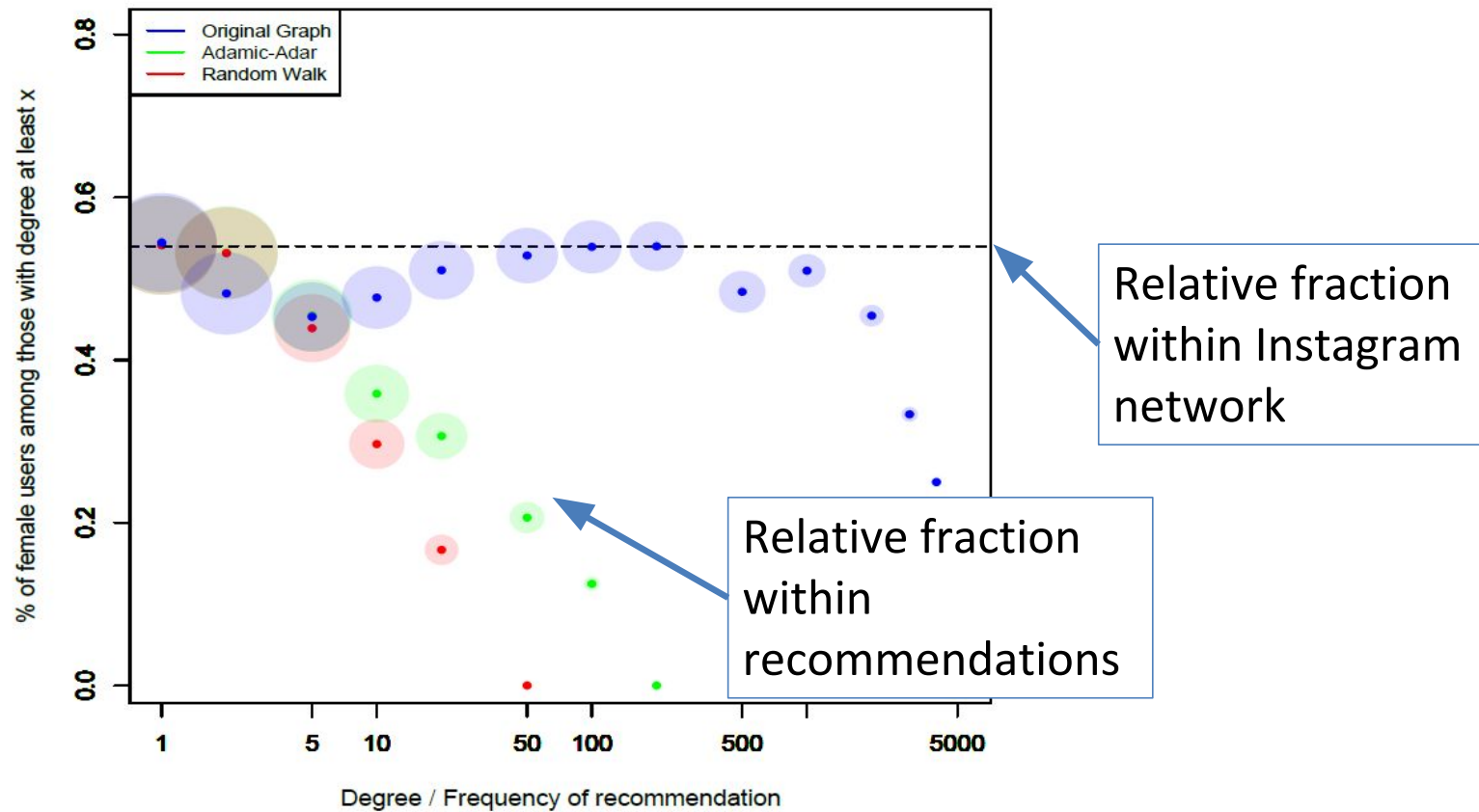
- Organic growth
  - While females are 54% of the population, they represent
  - 48% of users with degree  $> 10$  (top 1%)
  - 45% of users with degree  $> 2000$
- Recommendation
  - Use Adamic-Adar index to recommend new links
    - Suggests links to nodes with most common neighbors
  - Random walk recommendation
    - Choose a 2-hop neighbor at random

she didn't label but red is women and blue is men

## Bias under algorithmic link recommendation

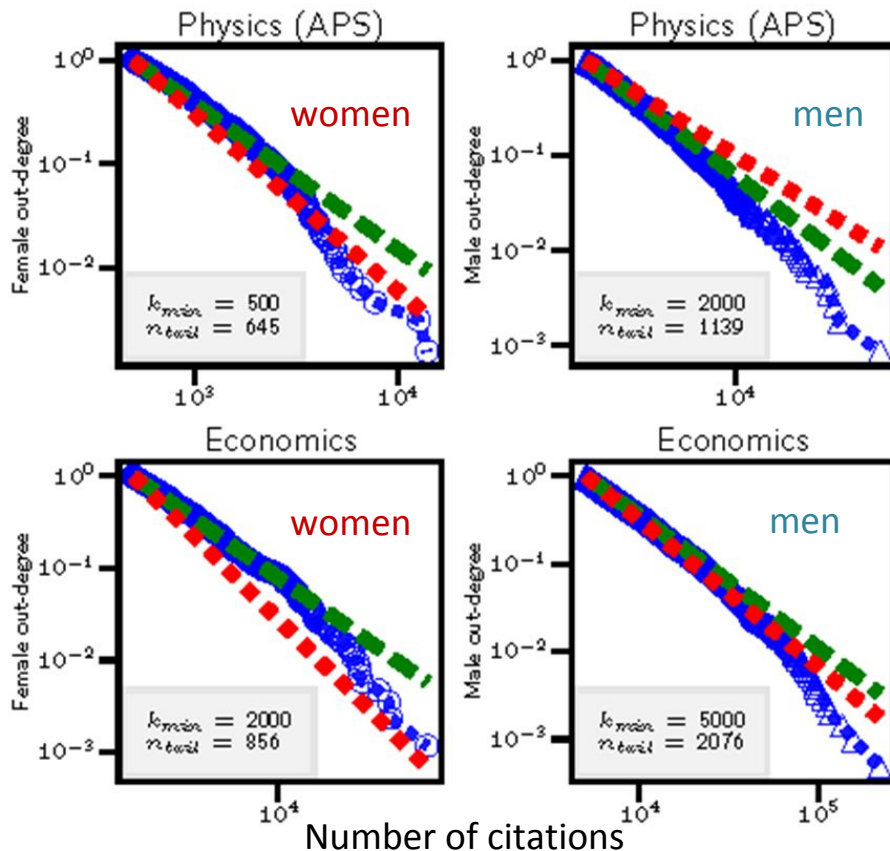


# Representation of women within top-recommendations





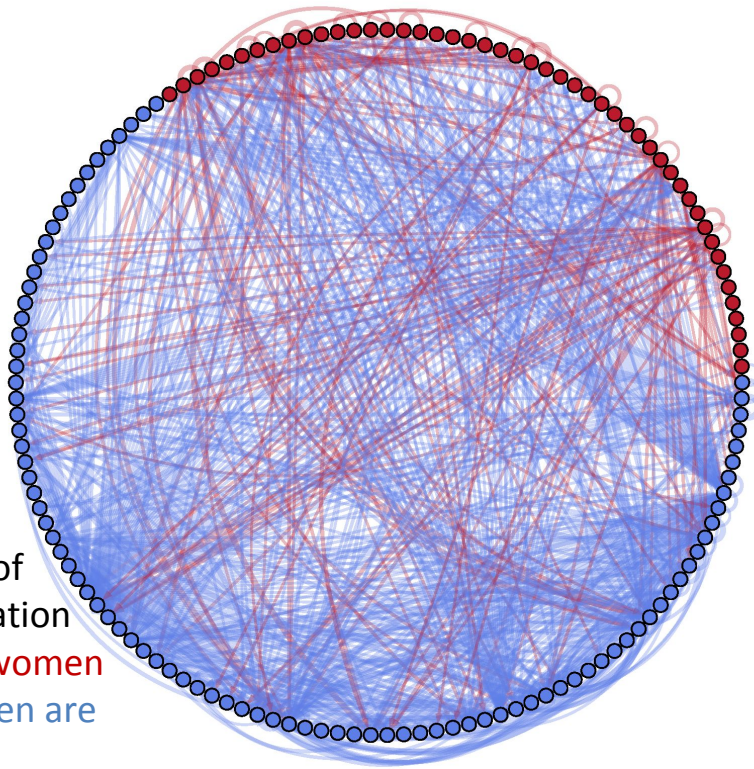
# Glass ceiling in science



- Number of citations is highly skewed
  - Most researchers have only a few citations, while a small number of researchers have large number of citations
- Citations are highly skewed for men
  - Top men researchers have more citations than top women researchers
  - Google scholar searches?



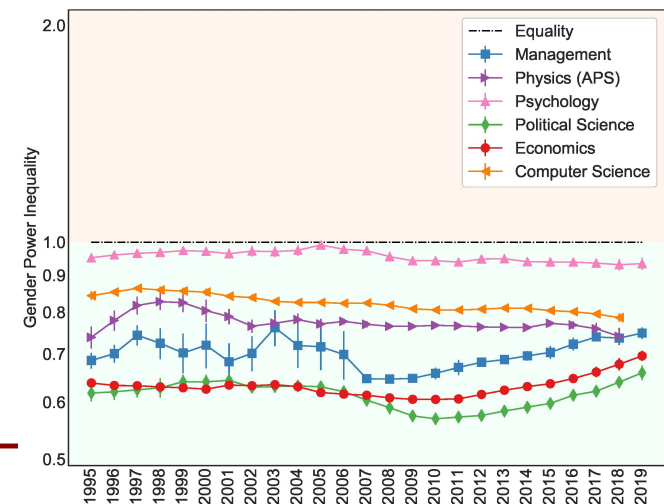
# Gender disparity in citations



Subgraph of  
author-citation  
network: **women**  
**are red**, **men are**  
**blue**

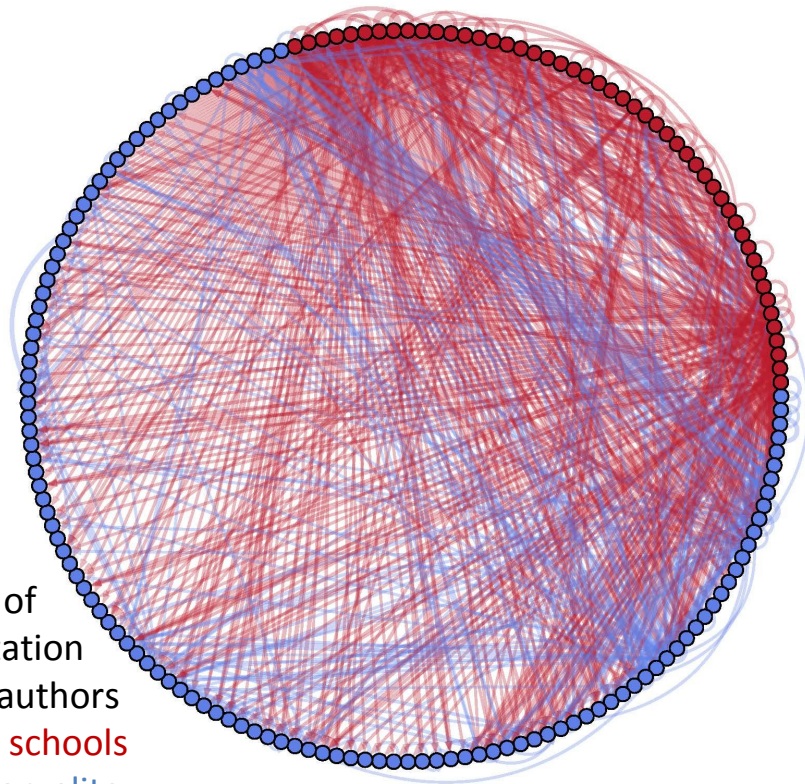
Source: <https://arxiv.org/abs/2103.10944>

- $d_o(v)$ : the number of authors who cite  $v$  (influence)
- $d_i(v)$ : the number of authors cited by  $v$
- Power inequality:
  - Which group has more influence: Red or Blue?
  - $I = d_o(\text{Red})/d_i(\text{Red}) * d_i(\text{Blue})/d_o(\text{Blue})$





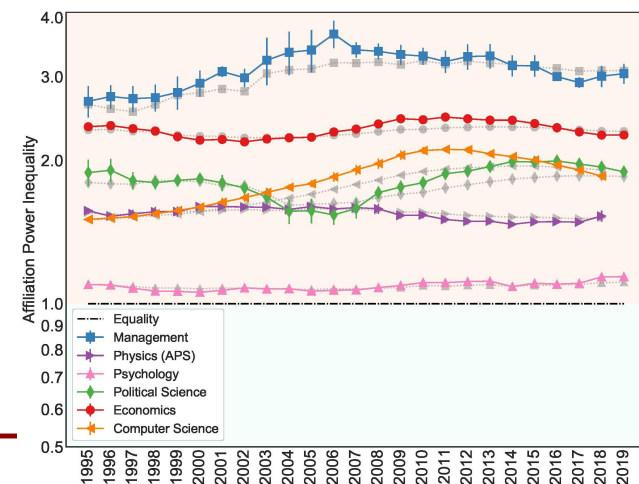
# Prestige bias in citations



Subgraph of  
author-citation  
network: authors  
from elite schools  
are red, non-elite  
are blue

Source: <https://arxiv.org/abs/2103.10944>

- $d_o(v)$ : the number of authors who cite  $v$  (influence)
- $d_i(v)$ : the number of authors cited by  $v$
- Power inequality:
  - Which group has more influence: Red or Blue?
  - $I = d_o(\text{Red})/d_i(\text{Red}) * d_i(\text{Blue})/d_o(\text{Blue})$

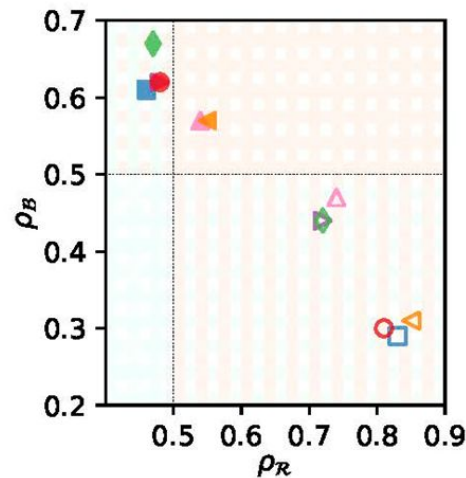




# Homophily creates citations disparity



Gender Disparity:  
Blue (men) are homophilic ( $\rho > 0.5$ ).  
Red (women) are mostly heterophilic.

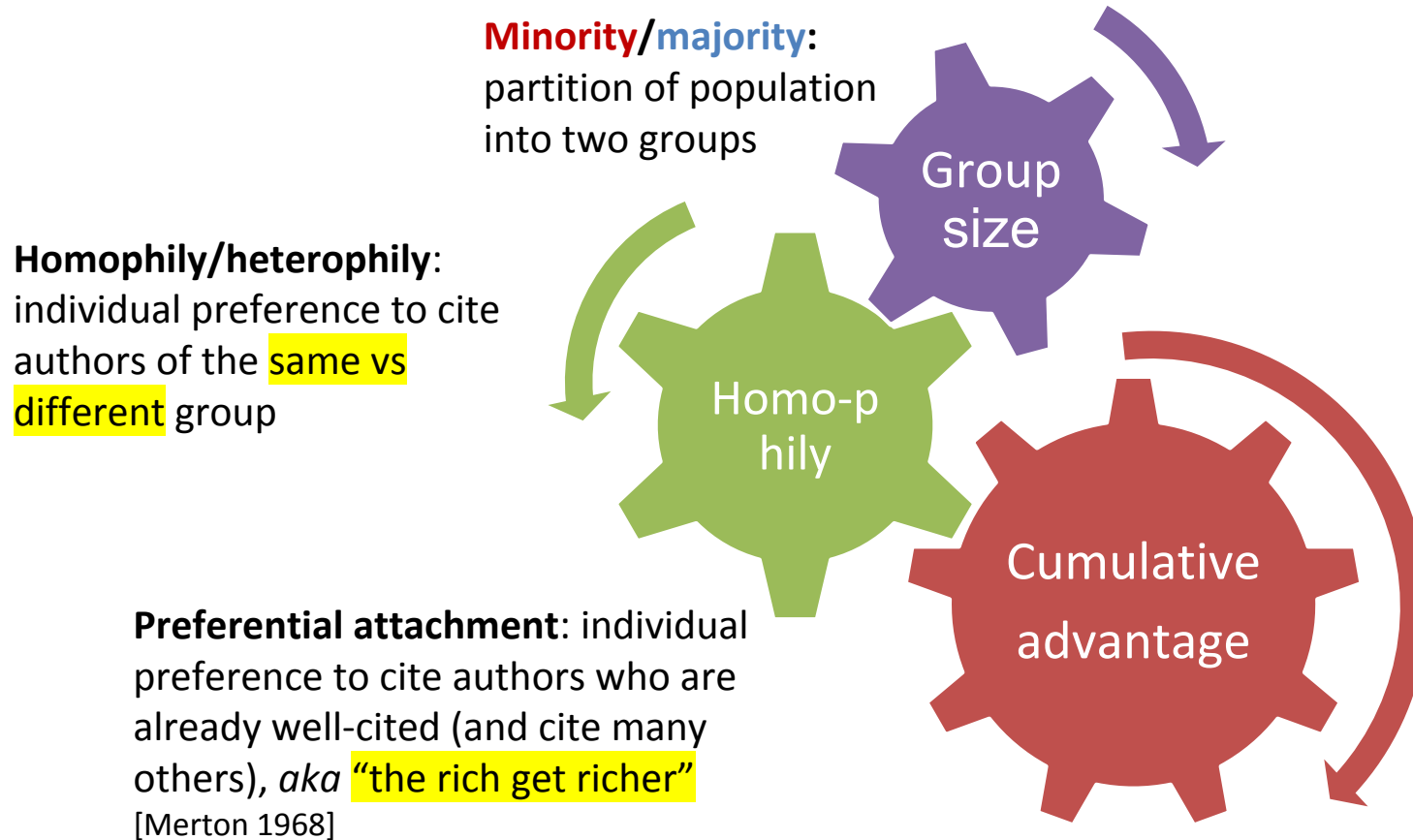


- Measure homophily from citations data
  - Preference of Red authors to cite other authors
  - $\rho_R > 0.5$  means Red authors prefer to cite other Red authors
  - $\rho_R < 0.5$  means Red authors prefer to cite Blue authors

Prestige Disparity:  
Red (elite institutions) are homophilic.  
Blue (non-elite) are heterophilic.



# Emergence of bias in citation networks





## Takeaways

- Homophily plays a central role in the glass ceiling effect
- Homophily creates advantages for some groups
- Reducing disadvantages in networks?